

JMO – 2003

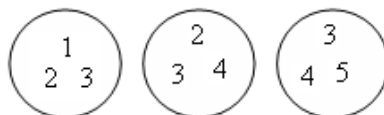
Max Mark – 100

Time – 3 Hrs

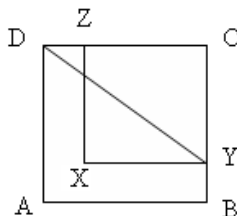
Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all questions.

1. Below there are three circle with some numbers inside each circle. Take only one number from one circle and keep it in another circle so that sum of numbers in each circles are equal.

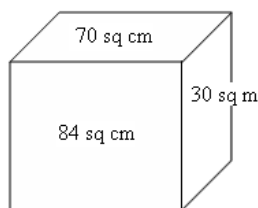


2. As per the BSNL to call abroad for each 13.6 sec it costs Rs. 30.90. Then what is the cost to call abroad for 3 minutes?
3. Arrange in ascending order $\frac{11}{15}, \frac{13}{19}, \frac{16}{23}$.
4. One shopkeeper sold two bicycles at Rs. 840 each. In one he got 40% profit and 25% loss in others. Then in total what is the % of loss or profit of shopkeeper?
5. Out of 50 colour balls in a pot, 13 are green, 10 are red, 9 are blue, 4 are white, 6 are black and 8 are yellow in colour. Without seeing the balls at least how many balls you can take so that it contains 7 balls of same colour?
6. Binod and Mary starts journey from their houses. If they travel by their own speed they will meet after 5 minutes. But if Binod starts his journey after 3 minutes traveled by Mary, then they will meet after 3 minutes. Find the time required to travel by Binod to Mary's house.
7. For an equal number of boys and girls asked to give answer YES or NO for a question "Do you love solving Maths problems?", if 70% are girls in giving answer YES and 80% are boys in giving answer NO. Then what is % of girls given answer YES?
8. In Pathani Samant Planetarium cost of one ticket for a child is Rs. 5 where as for adult it is Rs. 16. If in a day total collection is Rs. 789, then find at most number of peoples came to the planetarium.
9. What is the sum of digits of number we got after multiplication 999999×777777777 ?
10. Due to decrease in number of viewers in a cricket match, board of cricket decrease the cost of ticket by 20% for which in the next match selling of ticket increased by 20%. Then in totality in comparison to previous match what is the % of increase of decrease in earning from selling of tickets?
11. ABCD and XYCZ each are square. If Area outside XYCZ and area inside ABCD is 30 cm^2 and $DY = 10 \text{ cm}$, then find CD.

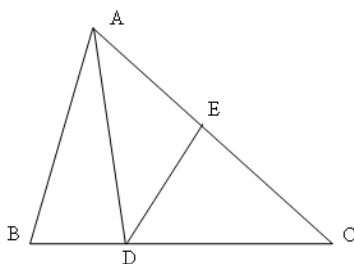


JMO QUESTIONS (ORISSA)

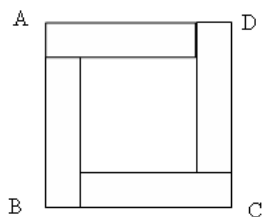
12. In the month of September in a town 20% affected by flu and 80% are healthy. In the month of October 20% people recovered from the flu affected people and 20% got affected from the healthy people. Then in the month of October end find the % of people are healthy.
13. Simplify: $1 + 2(1 + 2(1 + 2(1 + 2(1 + 2(1 + 2(1 + 2(1 + 2(1 + 2))))))))$.
14. If sum of 9 consecutive whole numbers is 180, then find the first number.
15. In the given cuboids each surface area are 84 cm^2 , 70 cm^2 and 30 cm^2 . Find the volume of cuboid. .



16. In the figure $AB = AC$, $m\angle BAD = 30^\circ$, $AE = AD$. Find $m\angle CDE = ?$



17. Four congruent rectangles mixed and form a square. If perimeter of each rectangle is 16 then find the area of square ABCD.



18. A679B is a five digit number which is divisible by 72 then find A and B.
19. Cost of 9 Mathematics and 16 Science books is Rs 220/-. If cost of each book is a whole number, then find cost of each Mathematics book.
20. Find smallest number divisible by 11, which when divided by 4, 5, 6 and 9 each time remainder is 1.

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